

CONSOLIDATED SOLUTION RANKING REPORT

Sensitivity Analysis with Weight Variation for Score and Coverage

This document presents the consolidation and structured analysis of sensitivity tests conducted on attribute solutions, evaluating the impact of weight variations on Score (predictive capacity/mutual information) and Coverage (representativeness/frequency) parameters.

The analysis is divided into two main blocks: the first scenario considers the original raw values, and the second block analyzes the scenario restricted to samples with a maximum coverage value equal to 12.

Block 1: Macro-Dense Scenario (Original Coverage Values)

In this scenario, the behavior of the solutions is evaluated without capping the sample ceiling of coverage, allowing the observation of which attribute combinations are intrinsically robust under different formulations of the combined utility function (Balanced Score).

1.1 Equal Weights (a = b = 0.5)

Pos	Attribute Solution	Coverage	Score	Balanced Score
1	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%.1-Passing']	0.4167	1.0000	0.7083
2	['90s-Defensive Actions', 'Cmp%.1-Passing', 'Cmp%-Passing']	0.5000	0.8975	0.6987
3	['90s-Defensive Actions', 'npG/Sh-Shooting', 'Cmp%.1-Passing']	0.4167	0.9639	0.6903
4	['90s-Defensive Actions', 'Cmp%.1-Passing', 'np:G-xG-Shooting']	0.4167	0.9281	0.6724
5	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%-Passing']	0.4167	0.9202	0.6684
6	['Fld-Goal and Shot Creation', '90s-Defensive Actions', 'npG/Sh-Shooting']	0.3333	0.9907	0.6620
7	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Lost-Miscellaneous Stats']	0.4167	0.8992	0.6579
8	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', '90s-Defensive Actions']	0.4167	0.8751	0.6459
9	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'np:G-xG-Shooting']	0.3333	0.9533	0.6433
10	['npG/Sh-Shooting', '90s-Defensive Actions', 'np:G-xG-Shooting']	0.3333	0.9192	0.6263
11	['npG/Sh-Shooting', '90s-Defensive Actions', 'Cmp%.1-Passing']	0.4167	0.8303	0.6235
12	['Dead-Pass Types', 'Cmp%.1-Passing', 'TI-Pass Types']	0.3750	0.8503	0.6126

Pos	Attribute Solution	Coverage	Score	Balanced Score
13	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'np:G/Sh-Shooting']	0.3333	0.8626	0.5980
14	['G/SoT-Shooting', 'Cmp%.1-Passing', 'np:G/Sh-Shooting']	0.2917	0.8786	0.5852
15	['90s-Defensive Actions', 'np:G-xG-Shooting', 'Fld-Goal and Shot Creation']	0.3333	0.8158	0.5746
16	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', 'G/SoT-Shooting']	0.2917	0.8522	0.5719
17	['Cmp%.1-Passing', 'np:G-xG-Shooting', 'G/SoT-Shooting']	0.2917	0.8427	0.5672
18	['G/SoT-Shooting', 'Fld-Goal and Shot Creation', 'np:G/Sh-Shooting']	0.2083	0.8911	0.5497
19	['G/SoT-Shooting', 'np:G-xG-Shooting', 'np:G/Sh-Shooting']	0.2083	0.8809	0.5446
20	['Fld-Goal and Shot Creation', 'G/SoT-Shooting', 'np:G-xG-Shooting']	0.2083	0.8532	0.5308

1.2 Heavier Score (a = 0.75, b = 0.25)

Prioritizing the internal statistical robustness of the score (individual and joint predictive quality) over the global frequency of coverage.

Pos	Attribute Solution	Coverage	Score	Balanced Score
1	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%.1-Passing']	0.4167	1.0000	0.8542
2	['90s-Defensive Actions', 'np:G/Sh-Shooting', 'Cmp%.1-Passing']	0.4167	0.9639	0.8271
3	['Fld-Goal and Shot Creation', '90s-Defensive Actions', 'np:G/Sh-Shooting']	0.3333	0.9907	0.8264
4	['90s-Defensive Actions', 'Cmp%.1-Passing', 'np:G-xG-Shooting']	0.4167	0.9281	0.8002
5	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'np:G-xG-Shooting']	0.3333	0.9533	0.7983
6	['90s-Defensive Actions', 'Cmp%.1-Passing', 'Cmp%-Passing']	0.5000	0.8975	0.7981
7	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%-Passing']	0.4167	0.9202	0.7943
8	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Lost-Miscellaneous Stats']	0.4167	0.8992	0.7786
9	['np:G/Sh-Shooting', '90s-Defensive Actions', 'np:G-xG-Shooting']	0.3333	0.9192	0.7727
10	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', '90s-Defensive Actions']	0.4167	0.8751	0.7605
11	['G/SoT-Shooting', 'Cmp%.1-Passing', 'np:G/Sh-Shooting']	0.2917	0.8786	0.7319

Pos	Attribute Solution	Coverage	Score	Balanced Score
12	['Dead-Pass Types', 'Cmp%.1-Passing', 'TI-Pass Types']	0.3750	0.8503	0.7315
13	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'npG/Sh-Shooting']	0.3333	0.8626	0.7303
14	['npG/Sh-Shooting', '90s-Defensive Actions', 'Cmp%.1-Passing']	0.4167	0.8303	0.7269
15	['G/SoT-Shooting', 'Fld-Goal and Shot Creation', 'npG/Sh-Shooting']	0.2083	0.8911	0.7204
16	['G/SoT-Shooting', 'np:G-xG-Shooting', 'npG/Sh-Shooting']	0.2083	0.8809	0.7128
17	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', 'G/SoT-Shooting']	0.2917	0.8522	0.7120
18	['Cmp%.1-Passing', 'np:G-xG-Shooting', 'G/SoT-Shooting']	0.2917	0.8427	0.7050
19	['90s-Defensive Actions', 'np:G-xG-Shooting', 'Fld-Goal and Shot Creation']	0.3333	0.8158	0.6952
20	['Fld-Goal and Shot Creation', 'G/SoT-Shooting', 'np:G-xG-Shooting']	0.2083	0.8532	0.6920

1.3 Heavier Coverage (a = 0.25, b = 0.75)

Scenario that favors the temporal and sectorial sample stability of attribute combinations.

Pos	Attribute Solution	Coverage	Score	Balanced Score
1	['90s-Defensive Actions', 'Cmp%.1-Passing', 'Cmp%-Passing']	0.5000	0.8975	0.5994
2	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%.1-Passing']	0.4167	1.0000	0.5625
3	['90s-Defensive Actions', 'npG/Sh-Shooting', 'Cmp%.1-Passing']	0.4167	0.9639	0.5535
4	['90s-Defensive Actions', 'Cmp%.1-Passing', 'np:G-xG-Shooting']	0.4167	0.9281	0.5445
5	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%-Passing']	0.4167	0.9202	0.5425
6	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Lost-Miscellaneous Stats']	0.4167	0.8992	0.5373
7	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', '90s-Defensive Actions']	0.4167	0.8751	0.5313
8	['npG/Sh-Shooting', '90s-Defensive Actions', 'Cmp%.1-Passing']	0.4167	0.8303	0.5201
9	['Fld-Goal and Shot Creation', '90s-Defensive Actions', 'npG/Sh-Shooting']	0.3333	0.9907	0.4977
10	['Dead-Pass Types', 'Cmp%.1-Passing', 'TI-Pass Types']	0.3750	0.8503	0.4938

Pos	Attribute Solution	Coverage	Score	Balanced Score
11	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'np:G-xG-Shooting']	0.3333	0.9533	0.4883
12	['np:G/Sh-Shooting', '90s-Defensive Actions', 'np:G-xG-Shooting']	0.3333	0.9192	0.4798
13	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'np:G/Sh-Shooting']	0.3333	0.8626	0.4656
14	['90s-Defensive Actions', 'np:G-xG-Shooting', 'Fld-Goal and Shot Creation']	0.3333	0.8158	0.4540
15	['G/SoT-Shooting', 'Cmp%.1-Passing', 'np:G/Sh-Shooting']	0.2917	0.8786	0.4384
16	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', 'G/SoT-Shooting']	0.2917	0.8522	0.4318
17	['Cmp%.1-Passing', 'np:G-xG-Shooting', 'G/SoT-Shooting']	0.2917	0.8427	0.4294
18	['G/SoT-Shooting', 'Fld-Goal and Shot Creation', 'np:G/Sh-Shooting']	0.2083	0.8911	0.3790
19	['G/SoT-Shooting', 'np:G-xG-Shooting', 'np:G/Sh-Shooting']	0.2083	0.8809	0.3765
20	['Fld-Goal and Shot Creation', 'G/SoT-Shooting', 'np:G-xG-Shooting']	0.2083	0.8532	0.3695

Block 2: Micro-Restricted Scenario (Sampling with Maximum Coverage Value = 12)

This block adjusts the coverage scale to a baseline where the maximum obtained value is considered a unitary ceiling (internal normalization per sample subgroup), reconfiguring the relative impact of coverage on the final synthetic index.

2.1 Equal Weights (a = b = 0.5)

Pos	Attribute Solution	Coverage	Score	Balanced Score
1	['90s-Defensive Actions', 'Cmp%.1-Passing', 'Cmp%-Passing']	1.0000	0.8975	0.9487
2	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%.1-Passing']	0.8333	1.0000	0.9167
3	['90s-Defensive Actions', 'np:G/Sh-Shooting', 'Cmp%.1-Passing']	0.8333	0.9639	0.8986
4	['90s-Defensive Actions', 'Cmp%.1-Passing', 'np:G-xG-Shooting']	0.8333	0.9281	0.8807
5	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%-Passing']	0.8333	0.9202	0.8767
6	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Lost-Miscellaneous Stats']	0.8333	0.8992	0.8663

Pos	Attribute Solution	Coverage	Score	Balanced Score
7	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', '90s-Defensive Actions']	0.8333	0.8751	0.8542
8	['npG/Sh-Shooting', '90s-Defensive Actions', 'Cmp%.1-Passing']	0.8333	0.8303	0.8318
9	['Fld-Goal and Shot Creation', '90s-Defensive Actions', 'npG/Sh-Shooting']	0.6667	0.9907	0.8287
10	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'np:G-xG-Shooting']	0.6667	0.9533	0.8100
11	['Dead-Pass Types', 'Cmp%.1-Passing', 'TI-Pass Types']	0.7500	0.8503	0.8001
12	['npG/Sh-Shooting', '90s-Defensive Actions', 'np:G-xG-Shooting']	0.6667	0.9192	0.7929
13	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'npG/Sh-Shooting']	0.6667	0.8626	0.7646
14	['90s-Defensive Actions', 'np:G-xG-Shooting', 'Fld-Goal and Shot Creation']	0.6667	0.8158	0.7412
15	['G/SoT-Shooting', 'Cmp%.1-Passing', 'npG/Sh-Shooting']	0.5833	0.8786	0.7310
16	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', 'G/SoT-Shooting']	0.5833	0.8522	0.7178
17	['Cmp%.1-Passing', 'np:G-xG-Shooting', 'G/SoT-Shooting']	0.5833	0.8427	0.7130
18	['G/SoT-Shooting', 'Fld-Goal and Shot Creation', 'npG/Sh-Shooting']	0.4167	0.8911	0.6539
19	['G/SoT-Shooting', 'np:G-xG-Shooting', 'npG/Sh-Shooting']	0.4167	0.8809	0.6488
20	['Fld-Goal and Shot Creation', 'G/SoT-Shooting', 'np:G-xG-Shooting']	0.4167	0.8532	0.6349

2.2 Heavier Score (a = 0.75, b = 0.25)

Pos	Attribute Solution	Coverage	Score	Balanced Score
1	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%.1-Passing']	0.8333	1.0000	0.9583
2	['90s-Defensive Actions', 'npG/Sh-Shooting', 'Cmp%.1-Passing']	0.8333	0.9639	0.9313
3	['90s-Defensive Actions', 'Cmp%.1-Passing', 'Cmp%-Passing']	1.0000	0.8975	0.9231
4	['Fld-Goal and Shot Creation', '90s-Defensive Actions', 'npG/Sh-Shooting']	0.6667	0.9907	0.9097
5	['90s-Defensive Actions', 'Cmp%.1-Passing', 'np:G-xG-Shooting']	0.8333	0.9281	0.9044
6	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%-Passing']	0.8333	0.9202	0.8985

Pos	Attribute Solution	Coverage	Score	Balanced Score
7	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Lost-Miscellaneous Stats']	0.8333	0.8992	0.8827
8	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'np:G-xG-Shooting']	0.6667	0.9533	0.8816
9	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', '90s-Defensive Actions']	0.8333	0.8751	0.8647
10	['np:G/Sh-Shooting', '90s-Defensive Actions', 'np:G-xG-Shooting']	0.6667	0.9192	0.8560
11	['np:G/Sh-Shooting', '90s-Defensive Actions', 'Cmp%.1-Passing']	0.8333	0.8303	0.8310
12	['Dead-Pass Types', 'Cmp%.1-Passing', 'TI-Pass Types']	0.7500	0.8503	0.8252
13	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'np:G/Sh-Shooting']	0.6667	0.8626	0.8136
14	['G/SoT-Shooting', 'Cmp%.1-Passing', 'np:G/Sh-Shooting']	0.5833	0.8786	0.8048
15	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', 'G/SoT-Shooting']	0.5833	0.8522	0.7850
16	['90s-Defensive Actions', 'np:G-xG-Shooting', 'Fld-Goal and Shot Creation']	0.6667	0.8158	0.7785
17	['Cmp%.1-Passing', 'np:G-xG-Shooting', 'G/SoT-Shooting']	0.5833	0.8427	0.7779
18	['G/SoT-Shooting', 'Fld-Goal and Shot Creation', 'np:G/Sh-Shooting']	0.4167	0.8911	0.7725
19	['G/SoT-Shooting', 'np:G-xG-Shooting', 'np:G/Sh-Shooting']	0.4167	0.8809	0.7649
20	['Fld-Goal and Shot Creation', 'G/SoT-Shooting', 'np:G-xG-Shooting']	0.4167	0.8532	0.7441

2.3 Heavier Coverage (a = 0.25, b = 0.75)

Pos	Attribute Solution	Coverage	Score	Balanced Score
1	['90s-Defensive Actions', 'Cmp%.1-Passing', 'Cmp%-Passing']	1.0000	0.8975	0.9744
2	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%.1-Passing']	0.8333	1.0000	0.8750
3	['90s-Defensive Actions', 'np:G/Sh-Shooting', 'Cmp%.1-Passing']	0.8333	0.9639	0.8660
4	['90s-Defensive Actions', 'Cmp%.1-Passing', 'np:G-xG-Shooting']	0.8333	0.9281	0.8570
5	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%-Passing']	0.8333	0.9202	0.8550
6	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Lost-Miscellaneous Stats']	0.8333	0.8992	0.8498

Pos	Attribute Solution	Coverage	Score	Balanced Score
7	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', '90s-Defensive Actions']	0.8333	0.8751	0.8438
8	['npG/Sh-Shooting', '90s-Defensive Actions', 'Cmp%.1-Passing']	0.8333	0.8303	0.8326
9	['Dead-Pass Types', 'Cmp%.1-Passing', 'TI-Pass Types']	0.7500	0.8503	0.7751
10	['Fld-Goal and Shot Creation', '90s-Defensive Actions', 'npG/Sh-Shooting']	0.6667	0.9907	0.7477
11	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'np:G-xG-Shooting']	0.6667	0.9533	0.7383
12	['npG/Sh-Shooting', '90s-Defensive Actions', 'np:G-xG-Shooting']	0.6667	0.9192	0.7298
13	['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'npG/Sh-Shooting']	0.6667	0.8626	0.7156
14	['90s-Defensive Actions', 'np:G-xG-Shooting', 'Fld-Goal and Shot Creation']	0.6667	0.8158	0.7040
15	['G/SoT-Shooting', 'Cmp%.1-Passing', 'npG/Sh-Shooting']	0.5833	0.8786	0.6572
16	['Fld-Goal and Shot Creation', 'Cmp%.1-Passing', 'G/SoT-Shooting']	0.5833	0.8522	0.6505
17	['Cmp%.1-Passing', 'np:G-xG-Shooting', 'G/SoT-Shooting']	0.5833	0.8427	0.6482
18	['G/SoT-Shooting', 'Fld-Goal and Shot Creation', 'npG/Sh-Shooting']	0.4167	0.8911	0.5353
19	['G/SoT-Shooting', 'np:G-xG-Shooting', 'npG/Sh-Shooting']	0.4167	0.8809	0.5327
20	['Fld-Goal and Shot Creation', 'G/SoT-Shooting', 'np:G-xG-Shooting']	0.4167	0.8532	0.5258

3. Conclusions and Methodological Guidelines

Based on multi-parameter sensitivity tests, critical patterns for selecting the final metric are identified:

- **Dominant Solution by Stability:** The attribute combination ['90s-Defensive Actions', 'Fld-Goal and Shot Creation', 'Cmp%.1-Passing'] shows exceptional resilience. It retains absolute leadership (1st place) in almost all sub-scenarios where the Score weight is predominant or balanced, recording a perfect Score ceiling of **1.0000**.
- **Inversion by Sample Ceiling:** When Coverage assumes the majority weight ($b = 0.75$), the trio containing ['90s-Defensive Actions', 'Cmp%.1-Passing', 'Cmp%-Passing'] moves to first place due to its larger linear coverage spectrum profile (Coverage = 0.5000 raw / 1.0000 micro-restricted), marginally sacrificing score quality (0.8975).
- **Academic Consistency Recommendation:** For scientific article publication purposes, it is recommended to adopt the balanced or score-biased model to maintain the conceptual robustness and mutual information that underpins the tactical role independence of the proposed metric.